

# Using Physiological Cues to Improve Empathy and Human–AI Interaction in a Mixed Reality Environment

COSC2148 / COSC3130 — Computing Research & Project Preparation

Assignment 3 — Research Proposal

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## 1 Introduction

Emotion is a fundamental dimension of human experience — a bounded, multicomponent phenomenon in which physiological changes, facial expressions, and bodily responses synchronise in response to events that are relevant to an individual’s needs and goals [1]. As human-computer interaction (HCI) becomes increasingly embedded in daily life, the field of affective computing has emerged to develop systems that detect human affect from observable signals and adapt their behaviour accordingly [2]. Physiological signals — including electroencephalography (EEG), electrodermal activity (EDA, also referred to as galvanic skin response), heart rate (HR), facial expressions, and eye gaze — have been established as reliable indicators of emotional states, each capturing distinct dimensions of the autonomic and somatic responses that accompany emotion [3–6].

Virtual Reality (VR), Augmented Reality (AR), and extended reality (XR) have emerged as immersive platforms for affective computing. Within this family, Mixed Reality (MR) — which blends real and virtual elements in a shared environment, following the reality-virtuality continuum of Milgram and Kishino [7] — has evolved from a theoretical taxonomy into an active research platform. Systematic reviews of empathic computing in immersive contexts confirm that physiological sensing [8] and AI-driven agent responses [9] have both been demonstrated in VR. MR, however, remains substantially underexplored as a context for empathic interaction between humans and AI [9].

A central requirement for empathic systems in immersive environments is personalisation. Empathy is engagement with an individual’s experience, not with a population average [10]; a system that responds to averages is not engaging empathically. The empirical record reinforces this requirement: the same emotion produces different physiological patterns in different individuals [11], and across VR and wearable contexts, user-independent classifiers consistently underperform personalised alternatives at affective classification by 15–30 percentage points in direct comparison [12, 13]. Individual calibration is therefore a design requirement rather than an enhancement [11]. Personalisation at the model level — fitting classifiers to each user’s physiological baseline — is now well established [14, 15]. What remains open is what those personalised signals mean for each user in context: physiological patterns can detect that a state is changing, but not what the change signifies for the individual in their current situation [1, 16]. Closing this interpretive layer is what distinguishes personalisation as a research claim from personalisation as a model-fitting exercise.

To close this interpretive gap, this project develops an empathic agent in Mixed Reality operating within a biocybernetic loop [11]: each user’s physiological signals — including electroencephalography, electrodermal activity, and eye gaze — are continuously sampled, classified into affective states, and used to adapt an LLM-based agent’s verbal response in real time [17]. The resulting interaction is evaluated through both objective physiological measures and subjective experiential instruments covering perceived empathy, trust, and presence. The work is structured around three research questions corresponding to detect–respond–align stages: how reliably real-time personalised physiological classification operates

during MR interaction (RQ1), whether continuous physiologically-adaptive agent response produces a measurably different user experience compared with a non-adaptive baseline (RQ2), and how objective physiological measures align with subjective experiential measures of empathy, trust, and presence (RQ3).

The proposal’s contribution to the academic literature lies in operationalizing an empathic mixed reality agent that bridges objective physiological measurement and subjective experiential measurement. This bridge is the methodological commitment that makes personalisation interpretive rather than merely model-level. Componential and contextual views of emotion [1, 16] hold that emotion is meaningful only as a joint bodily and experiential phenomenon: physiology captures automatic responses that participants may not consciously report, while subjective measures supply the contextual meaning and interpretation that physiology alone cannot. Existing empathic systems in extended reality have advanced one channel at a time. AffectivelyVR [14] personalizes emotion recognition from physiological signals and is evaluated through model performance; CAEVR [18] drives context-aware empathic responses from biosignals and reports physiological and subjective outcomes in parallel rather than integrated analysis. Chang’s work on empathic agents aware of user physiology [19] is evaluated primarily through subjective perception of agent empathy. The proposed system treats both channels as constitutive of how it understands and responds to emotion, and validates its interpretations through systematic analysis of their convergence and divergence. The contribution is therefore both a system design and an epistemological standard for empathic AI in MR: systems claiming to understand emotion should be evaluated against both bodily and experiential evidence. The remainder of this proposal develops these claims through three research questions, a literature review situating them in existing empathic computing work, a methodology operationalizing the design, and an evaluation plan.

## 2 Research Questions

This project develops an empathic Mixed Reality agent that combines personalised multimodal physiological classification with continuous adaptive response, evaluated across both objective and subjective measurements — an integration not yet demonstrated in MR. The three research questions below correspond to the system’s detection, response, and evaluation stages, each examined for its specific focus and significance.

### 2.1 RQ1 — Detection

*How reliably can real-time multimodal physiological classification, personalised through a brief pre-session calibration, transfer to infer users’ affective states during immersive Mixed Reality (MR) interaction?*

In seated, laboratory contexts, personalised models for physiological emotion classification have consistently outperformed user-independent alternatives at predicting predefined affective categories [12, 13, 15]. Whether these advantages survive the transfer to embodied MR interaction — where users move freely, wear an MR headset, and engage with both physical and virtual elements — is an open empirical question. The transfer is non-trivial: MR headsets occlude direct facial sensing [8], free movement introduces motion artefacts in EEG, EDA, and photoplethysmography (PPG) signals [18], and embodied interaction generates high-amplitude signal noise absent from controlled calibration [20]. RQ1 addresses this gap by testing whether per-participant calibration acquired in a controlled context, using validated affective stimuli, produces a classifier that remains reliable when the user is embodied, moving, and interacting in MR. Within this project the affective target is operationalised as a binary stress state, suited to the tense, exploratory MR scenario in which the system is tested. The significance is foundational: every downstream component of the system depends on reliable per-user signal detection in embodied MR. Answering RQ1 establishes whether personalisation transfers from calibration to interaction — and thereby whether the objective channel of the proposal’s bridge can support the work that follows.

### 2.2 RQ2 — Response

*Does an empathic AI agent that continuously adapts its verbal responses to users’ real-time physiological states produce a measurably different interaction experience compared with a non-adaptive baseline in Mixed Reality?*

Empathic agents that show awareness of users’ physiological states have been shown to increase perceived empathy, social connectedness, and trust in MR [19]. However, in Chang et al.’s study [19] this awareness was experimentally varied through programmatic conditions rather than driven by a real-time personalised classifier; the authors explicitly note that the impact of an agent’s imperfect detection on perceived empathy remains unclear. Closed-loop physiologically-adaptive systems have been demonstrated in adjacent contexts — adaptive environmental complexity in VR [21], adaptive game difficulty on desktop [22], and empathic conversational agents on screen-based digital humans [23] — but no system has connected real-time personalised classification with an LLM-based empathic agent in MR. Recent work on LLM-driven chatbots using nonverbal context [17] produced significant linguistic effects in response generation without significant differences in perceived empathy, reinforcing that whether classifier-driven adaptation produces measurable empathic effect is genuinely open. RQ2 addresses this gap by comparing the empathic Mixed Reality agent against a non-adaptive baseline matched for conversational surface — a comparison structure with precedent in biocybernetic loop methodology [11] and empathic interface agent evaluation [24]. The agent communicates through voice rather than text, consistent with voice being the dominant interaction modality in immersive agents for its alignment with real-time, embodied communication [25]. The significance is system-level: a measurable difference would demonstrate that personalisation from RQ1 translates, through continuous adaptive response, into empathically distinguishable interaction in MR; a null outcome, taken alongside Schmidmaier et al.’s similar finding for non-verbal-input LLM chatbots [17], would localise the limitation to either the integration architecture or to MR-specific factors. Either outcome bears on whether the bridge between detection and empathic effect holds in MR.

### 2.3 RQ3 — Alignment

*How do real-time physiological measures of affect align with users’ subjective experience of empathy, trust, and presence during empathic human-AI interaction in Mixed Reality, and what does analysis of their convergence and divergence reveal about empathy measurement in MR?*

Integrated analysis of physiological and subjective measures has established precedent in adjacent fields. Physiological computing routinely treats self-report as a route to concurrent validity for psychophysiological inference while recognising that the correspondence between channels is often erratic [11]. In empathy research focused on interactions between people, paired-measurement studies report context-dependent findings: physiological synchrony emerged but was not consistently associated with self-reported trait and state empathy measures in a study of 111 couples disclosing suffering [26], while EEG frontal asymmetry synchronisation was significantly associated with self-reported empathy in social VR with biofeedback [27]. Together, these findings indicate that the relationship between objective and subjective channels is real but variable. It emerges as an empirical question rather than an assumed property of valid measurement.

This analytical commitment has not been transferred to human-AI empathic agent research. A recent systematic review of empathic conversational agents in mental health care found that all studies conducting human evaluation of empathic features used self-reports alone [28]; Schmidmaier et al. [29], in developing the PETS scale, note that no validated comparator measures exist against which the concurrent validity of perceived system empathy can be tested. Among empathic agent studies in MR and VR, AffectivelyVR [14] reports model performance only, CAEVR [18] collects both channels but analyses them in parallel pipelines, and Chang et al. [19] use physiological data as input while reporting only subjective outcomes. None performs the integrated analysis.

Emotion is meaningful only as a joint bodily and experiential phenomenon [1,16]: physiological signals indicate state changes, but only the user’s reported experience supplies what the change means in context. For an empathic system whose contribution depends on engaging with an individual’s experience rather than a population average [10], convergence between channels is the evidence that the system is doing what it claims; divergence is a diagnostic finding about where the system’s interpretation breaks down. RQ3 transfers this analytical commitment into human-AI interaction in MR, analysing how objective physiological measures and subjective measures, covering empathy, trust, and presence, converge and diverge across the same participants and the same interactions. The significance is epistemological: a framework that treats convergence as supporting evidence and divergence as diagnostic information contributes a methodological standard for evaluating empathic AI in MR.

### 3 Literature Review

This literature review draws on prior research to identify the gap this project addresses, tracing six threads: the foundations of emotion, empathy, and physiological sensing; the emergence of immersive environments for empathic computing; the personalisation that physiological detection requires; the progression of empathic agents from rule-based to LLM-driven response; the closed-loop systems that adapt to physiological state continuously; and the measurement of empathy in human-AI interaction.

#### 3.1 Foundations: Emotion, Empathy, and Physiological Sensing

Emotion is represented in affective research through three broad traditions [6]. Categorical models treat emotion as a set of discrete, cross-culturally recognisable states such as happiness, fear, or anger [4]; dimensional models place affect in a continuous low-dimensional space, of which Russell’s circumplex [16] — locating any state by its valence (pleasant-unpleasant) and arousal (activation-deactivation) — is the most widely used [8]; and componential approaches treat emotion as a process, with Scherer [1] modelling it as an integrated pattern across appraisal, physiological response, expression, and subjective feeling. Because physiological response is among these components, physiological signals provide a candidate channel for emotion recognition. Most physiological emotion recognition operationalises affect in the valence-arousal terms of the dimensional model [6, 16].

Where emotion concerns an individual’s own affective state, empathy concerns responding to the state of another. Cuff et al. [30], reviewing 43 definitions, characterise empathy as a multidimensional construct combining a cognitive component (understanding another’s state) and an affective component (sharing or responding to it), and distinguish it from related notions such as sympathy and emotional contagion — the latter separated from empathy by the observer’s awareness that the felt emotion originates in another. Preston and de Waal [10] ground this in the Perception-Action Model, in which perceiving another’s state activates the observer’s own representations of it — an account that locates empathy in engagement with a specific individual rather than a population average. For an artificial system, empathy is correspondingly understood as responsiveness to a particular user’s emotional state [31], which frames both the design problem of detecting and responding to that state and the measurement problem of assessing perceived empathy [29].

The premise that physiological signals can index emotion is the foundation of affective computing, the field Picard [2] established on the proposition that machines could recognise and respond to human emotion through measurable signals. These signals fall into two broad groups: central-nervous-system activity, captured primarily by electroencephalography (EEG), and peripheral activity [3], including electrodermal activity (EDA), cardiac measures such as heart rate and heart rate variability, and oculomotor measures such as gaze and pupil response [5]. Reviews of the field document that these signals carry emotion-relevant information across a range of tasks and populations, providing a continuous and objective channel that complements self-report and outward behaviour [3, 6].

#### 3.2 Immersive Environments for Empathic Computing

Mixed Reality (MR) occupies the middle of the reality-virtuality continuum defined by Milgram and Kishino [7]: where Virtual Reality immerses the user in a wholly synthetic world, MR merges real and virtual content within a single shared display, with Augmented Reality its best-known instance. This blend makes MR a distinct setting for embodied interaction — the user engages a virtual agent while remaining present in, and able to act within, the physical environment.

A PRISMA systematic review of immersive technologies for empathic computing from 2000 to 2024, organised around emotion elicitation, emotion recognition, and fostering empathy, finds that VR has been the dominant platform for two decades while AR and MR remain underexplored [9]; dimensional emotion models have shaped this work more than discrete ones [9]. The same concentration appears in embodied conversational agents, where VR-based head-mounted implementations dominate the XR literature [25].

Within these immersive agent systems, physiological sensing is rarely integrated. Across twenty-three XR embodied-agent studies, only one incorporated a physiological signal (heart rate), and none used multimodal physiological classification of the user’s emotional state [25]. The setting that combines MR with physiologically-aware empathic interaction is therefore doubly underexplored — both as an

environment, since MR empathic computing itself is sparse [9], and as an input modality, since immersive agents seldom sense the user’s physiological state at all [25].

### 3.3 Personalisation in Physiological Emotion Classification

Physiological signals offer a route to emotion recognition that is less susceptible to conscious control than behavioural cues such as facial expression or speech, but their interpretation is complicated by individual variability [3]. Fairclough [11] identifies the reliability of psychophysiological inference across individuals as a fundamental design challenge for physiological computing. Reviews of physiological emotion recognition report that this variability — differing signal baselines and emotional physiologies across people — leaves subject-independent models performing well below subject-dependent ones and not yet reliable for real-time use [3]; in EEG-based recognition specifically, individual differences in neurophysiology and baseline activity make a single cross-subject model difficult to achieve, motivating adaptive and personalised approaches [32].

This pattern holds across measurement contexts. In non-immersive settings, Li and Washington [13], classifying physiological signals against the induction condition (neutral, stress, or amusement) in the wearable-based WESAD dataset, found personalised models reached approximately 95% accuracy against approximately 67% for generalised models; Fu et al. [15], classifying signals against participants’ self-rated valence and arousal on the DEAP dataset, attributed the cross-individual gap to distributional shifts between users and applied transfer learning to adapt across them. The pattern is sharper in immersive settings: Lima et al. [12], classifying physiological signals against the predefined emotional category of film stimuli (drawn from the EMDB) during VR viewing, found user-dependent classifiers reached 71% accuracy while user-independent classifiers reached only 24.9%, concluding that immersive environments produce higher response variability that hardens generalisation, and AffectivelyVR [14] reported 96.5% accuracy for a per-participant EEG and EDA model against 83.7% for a generalised one when classifying self-rated valence during 360-degree video viewing.

Two features of this body of work bear on its application to Mixed Reality. First, personalisation is calibrated under controlled conditions — typically seated viewing of standardised stimuli — and evaluated within that same context [12, 14]. Yet applying laboratory-calibrated models to less constrained settings is known to degrade performance, as environmental noise, distraction, and variable engagement reduce classification reliability outside the lab [32]. Second, embodied MR interaction introduces signal noise that seated calibration does not: body movement and headset pressure generate high-amplitude EEG artifacts that can force the removal of affected channels [20], and concurrent speech introduces conversational noise that has been found to mask EEG differences between conditions [23]. Whether a classifier personalised through pre-session calibration transfers to affective classification during active, embodied MR interaction therefore remains an open question.

### 3.4 Empathic Agents: From Rule-Based to LLM-Driven

Empathic agents respond to a user’s inferred emotional state through facial expression, behaviour, or verbal communication. Prendinger et al. [24] demonstrated an early example in a Skip-Bo card game, where an animated agent modulated its behavioural and verbal responses to the user’s affective state, using physiological signals (skin conductance, electromyography) alongside self-report to evaluate whether the empathic condition produced detectable differences in user experience. In immersive XR contexts, voice-based interaction is the dominant modality for empathic agents, valued for its alignment with real-time communication and its capacity to strengthen emotional resonance when synchronised with nonverbal cues; the mechanism for generating that verbal response has evolved from rule-based templates toward generative language models [25].

Within these settings, empathic agents have used rule-based or programmatic response generation. CAEVR [18] combined real-time emotion recognition (from EEG, EDA, and HRV) with a context-aware empathic virtual agent that delivered predefined dialogue through a Belief-Desire-Intention framework; the empathic agent significantly increased perceived empathy, while an emotion-adaptive environment alone did not — locating the empathic effect in the agent’s verbal response rather than environmental adaptation. In MR, Chang et al. [19] demonstrated that an agent’s verbal acknowledgment of user physiological state — varied through programmatic conditions — significantly increased perceived empathy,

social connectedness, and trust over a non-adaptive baseline. On screen-based platforms, Saffaryazdi et al. [23] developed an empathic conversational agent for a digital human, classifying user emotion from EEG, EDA, and PPG every five seconds and mirroring it through facial expression and predefined verbal response. Both CAEVR and Saffaryazdi explicitly identify LLM-based response generation as the next step their rule-based implementations did not reach.

Large language models replace predefined templates with generated conversational responses, but their empathic capacity has limits. Schmidmaier et al. [17] found that multimodal LLMs conditioned on user nonverbal cues produced significant linguistic differences in generated utterances yet no significant difference in perceived empathy. Dongre et al. [31] proposed an EmLLM framework integrating multimodal physiological sensing with LLM response, demonstrated through a stress-management chatbot achieving 85% stress detection accuracy from PPG, EDA, and skin temperature.

The field can be located along three positions. In MR, empathic agents have shown that awareness of user physiological state increases perceived empathy — but this awareness has been driven by programmatic conditions rather than continuous classification [19]. In adjacent contexts — VR, screen-based digital humans, and chatbots — empathic agents have integrated real-time physiological classification [18, 23] and, separately, LLM-driven verbal generation [17, 31], but not together in an immersive setting. What remains undeveloped is their integration: a Mixed Reality empathic agent whose verbal, LLM-driven responses are continuously adapted by personalised real-time physiological classification of the user’s state.

### 3.5 Closed-Loop Adaptive Systems and the Biocybernetic Loop

Continuous physiological adaptation is formalised in the biocybernetic loop, which Fairclough [11] describes as a control cycle that translates physiological data into system adaptation: signals are sampled, an inference about the user’s state is drawn, the system adjusts in response, and the changed state is sampled again. Decision rules link inferred states to adaptations [11], distinguishing this continuous, looped operation from a single classification performed once.

This loop has been closed successfully in adjacent domains, predominantly around environmental or task parameters. Ewing et al. [22] used EEG measures as inputs to a biocybernetic loop that adjusted game difficulty to sustain player engagement, following the motivational intensity model. Chiossi et al. [21] implemented a closed-loop VR system that adapted the visual complexity of the environment to the user’s electrodermal arousal, improving working-memory performance relative to a non-adaptive baseline. Oner and Kucukkulahli [33] drove dynamic difficulty adjustment in a survival-horror game from real-time heart-rate and stress signals on a "relax-to-win" principle, raising intrinsic motivation while reducing physiological arousal.

In each case the loop adapts the environment or the task, not the agent’s communication; the established approach in physiologically-adaptive immersive systems has been dynamic adjustment of task difficulty or environmental parameters [21]. Closing the loop around an empathic agent’s continuous verbal response — sampling the user’s state, classifying it, adapting what the agent says, and re-sampling as the user responds — has not been demonstrated, in Mixed Reality or otherwise; immersive embodied agents seldom sense the user’s physiological state at all [25]. Extending the biocybernetic loop from environmental adaptation to the continuous verbal response of an empathic MR agent therefore remains an open problem.

### 3.6 Empathy Measurement and Cross-Channel Validation

How empathy is measured in human-AI interaction (HAI) reflects a methodological narrowness. Empathy from an artificial system toward a user is most commonly assessed through self-report scales — instruments adapted from human empathy research (the Interpersonal Reactivity Index, Toronto Empathy Questionnaire) and instruments designed specifically for systems (the Robot’s Perceived Empathy scale, the Empathy Scale for Human-Computer Communication) [29]. The Perceived Empathy of Technology Scale (PETS), developed by Schmidmaier et al. through multi-stage validation across more than 700 participants, was designed to address methodological limitations in earlier instruments — yet its developers explicitly note that no validated comparator measures exist against which the concurrent validity of perceived system empathy can be tested [29]. A recent systematic review of 19 empathic conversational

agent studies in mental health care confirms the pattern across the field: all five studies that conducted human evaluation of empathic features used self-reports alone [28]. The methodological norm in HAI is to ask users whether they perceive empathy and to triangulate this answer against no other source of evidence [28, 29].

In adjacent fields, the analytical move of pairing subjective measures with objective evidence is well-established. In physiological computing, Fairclough [11] recommends self-report as a route to concurrent validity for psychophysiological inference, while noting that the correspondence between these channels is often erratic — a tension he traces to the asymmetry between self-report (a conscious act of introspection) and psychophysiological response (reflecting both conscious and unconscious processes). In interpersonal empathy research, paired-measurement studies report context-dependent findings. In a study of 111 couples disclosing experiences of suffering, physiological synchrony emerged in skin conductance and interbeat interval reactivity but was not consistently associated with self-reported trait or state empathy measures [26]. In social VR with biofeedback, EEG frontal asymmetry synchronisation was significantly associated with self-reported empathy [27]. Together these findings suggest that the relationship between objective and subjective channels of empathic experience is real but variable. Measurement context — the task, the relational setting, the presence or absence of biofeedback — appears to modulate when channels align [26, 27].

In human-AI empathic interaction, this analytical commitment has not been transferred. The systems closest to integrated evaluation in immersive contexts treat the two channels in parallel rather than as relational evidence about empathic experience. AffectivelyVR [14] reports model performance metrics only; CAEVR [18] collects both physiological signals (EDA, HRV, EEG features) and multiple subjective scales (covering empathy, flow, and workload) but analyses them through separate statistical pipelines; Chang et al. [19] use physiological data as input to determine agent awareness conditions while reporting only subjective social perception outcomes. The absence of cross-channel analysis in HAI research is not an oversight; it reflects the absence of a methodological framework specifying what convergence between channels would constitute evidence of and what divergence would diagnose. For systems claiming to be empathic — claims that by their nature address an individual user’s experience rather than population averages [10] — establishing such a framework remains a methodological gap whose closure has not been attempted in MR.

## 4 Research Methodology

This project follows a design-based research (DBR) methodology [34], in which a system is refined across iterative cycles that each test and revise a design conjecture rather than tune parameters in isolation. The methodology rests on a componential, contextual view of emotion [1, 16]: physiological signals register automatic responses that the user may not consciously report, while subjective experience supplies the contextual meaning that physiology alone cannot. This commitment shapes the design throughout — detection is calibrated objectively from stimulus-labelled physiology, while subjective experience is held back as an independent channel against which the system’s interpretations are tested. The system is built around a personalised binary stress classifier driving an adaptive LLM-based agent, and the methodology is organised by the project’s detect–respond–align spine: each research question is addressed in turn, and the early iterations test the detection and response conjectures at small scale before the main study evaluates them in full.

### 4.1 Methodological approach and task design

The study uses a within-participants design in which every participant completes two phases: a controlled calibration phase and an embodied interaction task. The two are deliberately distinct contexts, and whether the classifier transfers across that gap is what RQ1 tests. The participant sample is determined by the power analysis in §5.

**Calibration** is seated and controlled. Validated stress-inducing stimuli — short tense audiovisual clips contrasted against calm, neutral clips — supply stimulus-derived labels for the binary stress / non-stress target defined in §4.2, with a brief self-rating after each clip confirming that the intended state was induced for that participant. The clips are drawn from a validated affective film database

with a dedicated high-arousal aversive category, such as the Emotional Movie Database (EMDB) used for immersive emotion classification in the work reviewed in §3.3 [12], fixing the stimulus class as a reproducible entry condition while leaving the final clip selection to be confirmed in the technical pilot. Pre-rated affective clips are the established calibration paradigm in the reference datasets and the broader emotion-recognition literature [6, 35, 36]. This phase fits the per-participant classifier on clean, low-movement signal.

**The interaction task** is an exploratory Mixed Reality scenario built on sustained ambient tension rather than discrete startle events. This design elicits a sustained stress state, which suits continuous classification and gives the agent an ongoing state to respond to rather than momentary reflexes to react against. The participant freely explores a dimly lit, tense virtual environment rendered over a controlled laboratory space, pursuing a loose objective (locating and reaching a target point). Conducting the study in a fixed laboratory setting holds the physical environment constant across participants, supporting internal validity; the objective complements this by standardising movement and exposure, preventing disengagement and keeping behavioural conditions comparable. Throughout, an empathic agent accompanies the participant as a voiced companion, offering reassurance and commentary; voice is the dominant modality for embodied agents for its alignment with real-time, embodied interaction [25]. This task is the source of all RQ2 and RQ3 data, and its embodied conditions — headset worn, free movement, active interaction — are precisely what the classifier calibrated in the controlled phase must contend with.

## 4.2 Addressing RQ1 — Detection

RQ1 asks how reliably a personalised multimodal classifier, calibrated in the controlled context of §4.1, transfers to detect user stress during the embodied interaction task. The system captures electroencephalography (EEG), electrodermal activity (EDA), and eye gaze throughout each session. Table 1 sets out the sensing capability each modality requires and example research-grade instruments that meet it; the requirements are fixed by the method, while the specific devices are selected during the technical pilot (§4.5), where matching sensors to the MR form factor and to acceptable signal quality under movement is itself a design question. The example configurations are drawn from comparable empathic-agent studies in MR and VR, demonstrating that the modality set is realisable on existing instruments [18, 19].

Table 1: Sensing requirements per modality, with example instruments. Requirements are method-driven; devices are illustrative and selected in the technical pilot.

| Modality   | Required capability                                                                                              | Example instrument(s)                                             |
|------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| EEG        | Multi-channel scalp EEG, frontal/prefrontal coverage, $\geq 128$ Hz, raw stream accessible                       | OpenBCI EEG cap; Emotiv; Muse                                     |
| EDA        | Tonic (SCL) and phasic (SCR) skin conductance, continuous sampling, raw stream accessible                        | Shimmer3 GSR+; Empatica                                           |
| Eye gaze   | Fixation and pupillary measures with the <i>raw gaze stream exposed to the application</i> (not gaze-to-UI only) | HMD-integrated trackers (e.g. Vive Pro Eye, Quest Pro); Tobii Pro |
| MR display | Standalone passthrough MR headset supporting the rendered tense environment                                      | Meta Quest 3; HTC Vive (with passthrough)                         |

The calibration phase supplies stimulus-derived labels for a **binary stress** / **non-stress** target. Stress is selected as the operational target for two reasons. First, horror-based stress induction is an established paradigm for which binary stressed/non-stressed classification has been demonstrated [37]. Second, electrodermal activity reflects sympathetic arousal directly, making an arousal-aligned target the construct the chosen signals measure most reliably [6, 12]. A binary target also avoids committing the classifier to valence regions of the affective space that a uniformly tense scenario does not populate.

Labelling from the stimulus rather than from self-report keeps detection objective and reserves subjective ratings for the independent convergence analysis in RQ3.

Personalisation operates at the model level: a classifier is fitted per participant, and a short baseline acquisition inside the MR environment without affective content provides a per-participant reference point for normalising classifier output during the main interaction. Classification uses classical supervised models — support vector machines and random forests — over engineered features standard for each modality: tonic skin conductance level and phasic skin conductance response measures for EDA, band-power spectral features for EEG, and fixation and pupillary measures for gaze [3,12]. The three modalities are combined by feature-level fusion into a single per-participant feature vector [36]. Classical models with engineered multimodal features are appropriate to the data regime: per-participant calibration produces small training sets, for which these models are better matched than deep architectures that require substantially more data, and they are the established choice in comparable immersive emotion-classification work [12,14]. The final model is selected in the technical pilot by cross-validated F1 — the appropriate metric given the class imbalance inherent in stress detection, where stressed episodes are the minority class — with model selection detailed in §4.5. The primary answer to RQ1 is the classifier’s stress-detection F1 during the interaction task relative to its calibration performance, quantifying whether personalisation transfers from the controlled context to embodied interaction.

### 4.3 Addressing RQ2 — Response

RQ2 asks whether a physiologically-adaptive empathic agent produces a different interaction experience from a non-adaptive baseline in MR. Prior physiologically-adaptive systems have established the comparison structure this question requires — an adaptive condition tested against a matched non-adaptive control, analysed within participants [21] — and empathic interface agents responding to user physiology have been evaluated against non-adaptive equivalents [24]. Within the interaction task of §4.1, each participant experiences two agent conditions: an LLM-based agent whose verbal responses are conditioned in real time on the RQ1 classifier’s stress output, forming a closed biocybernetic loop [11,17,31], and a non-adaptive baseline matched for conversational surface but independent of physiological state [21]. The baseline uses the same LLM and voice, with the stress-state context field withheld from its prompt, so the two conditions differ only in whether physiological state drives the response — holding utterance frequency, length, and conversational role constant. The initial conditioning mechanism injects the classifier’s current stress state into the agent’s prompt as a structured context field that the system prompt instructs the model to respond to, following the prompt-augmentation approach demonstrated for affective state in prior LLM agents [17]; the precise mapping from state to response and its timing are what Cycle 2 refines (§4.5). Condition order is counterbalanced across participants. The primary answer to RQ2 is the within-participant difference in subjective interaction experience between the two conditions (instruments and analysis in §5).

### 4.4 Addressing RQ3 — Alignment

RQ3 asks how real-time physiological measures align with users’ subjective experience during empathic interaction in MR. Prior empathic-agent studies report physiological and subjective measures separately, or only one of the two, with no integrated analysis in MR [18,19]. Because the classifier is trained on stimulus-derived labels rather than on the self-reports it is later compared against, the alignment between the two channels remains an independent empirical question rather than a built-in property of the design. For the same participants and the same interactions, the project relates real-time classifier output to post-session subjective ratings covering empathy, trust, and presence, analysing their convergence and divergence (procedure in §5).

### 4.5 Iteration cycles

The methodology is operationalised through two refinement cycles preceding the main study, each testing a design conjecture central to a research question by observing a defined measure, revising the design where the measure falls short, and re-testing until an exit condition is met (Table 2). **Cycle 1 (technical pilot)** tests whether per-participant calibration transfers to embodied MR. Although the laboratory

setting holds the environment constant, the participant still moves and explores freely while wearing the headset, so MR introduces signal artefacts absent from seated calibration — motion and embodied interaction generating high-amplitude noise in EEG, EDA, and gaze [18, 20]. The cycle therefore establishes artefact-handling (filtering, baseline normalisation, and artefact rejection) as a precondition for reliable detection, and selects the classifier model by cross-validated F1. **Cycle 2 (process pilot)** then tests whether classifier-driven adaptation produces a perceptibly different interaction, refining the mapping from classifier state to agent response and the interaction timing on a small number of pilot users. The two cycles are sequential rather than parallel: Cycle 2 depends on the detection pipeline that Cycle 1 produces, since the agent cannot be adapted to a stress signal that is not yet reliably detected — so a weak transfer result in Cycle 1 is surfaced and addressed before the response mapping is piloted, rather than confounding the main study. The main study then evaluates the frozen system across all participants. This structure ties each iteration to a design principle under test rather than to incremental tuning, and aligns the cycles with the detect–respond stages they refine.

Table 2: Design-based research cycles: each tests a conjecture against a defined measure and exit condition before the main study.

| Cycle                             | Conjecture tested                                                                      | What it refines                                                                                                            | Exit condition                                                                                                                  |
|-----------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Cycle 1 —<br>technical pi-<br>lot | Per-participant cali-<br>bration transfers to<br>embodied MR (RQ1)                     | Sensing pipeline, feature<br>set, artefact-handling pa-<br>rameters; classifier model<br>selected by cross-validated<br>F1 | Interaction-task F1<br>clears the majority-<br>class floor and<br>the calibration-to-<br>interaction transfer gap<br>stabilises |
| Cycle 2 —<br>process pilot        | Classifier-driven<br>adaptation yields a<br>perceptibly different<br>interaction (RQ2) | State-to-response map-<br>ping, response timing,<br>and conversational-surface<br>match between conditions                 | Adaptive condition<br>reads as coherently re-<br>sponsive and is cleanly<br>distinguishable from<br>the matched baseline        |

## 5 Evaluation

Evaluation addresses the three research questions through complementary measurement strategies: objective classifier performance for RQ1, subjective interaction experience for RQ2, and the integrated analysis of both for RQ3. The confirmatory hypothesis test is the RQ2 within-participant comparison of the adaptive and non-adaptive conditions, and the sample size is set by powering this test. Using a Wilcoxon signed-rank test at  $\alpha = 0.05$  and 80% power, with the effect size drawn from the closest precedent — Chang et al.’s within-subjects Mixed Reality study, in which the contrast between a physiology-aware and a physiology-unaware agent produced moderate effects on perceived empathy and related constructs [19] — yields a target sample in the low-to-mid twenties, consistent with comparable MR and VR empathy studies [18, 19] and justified following current sample-size guidance [38]. The study is therefore powered to detect a moderate or larger effect; a small effect would require a substantially larger sample than is feasible here, a limitation made explicit rather than assumed away. Every participant completes both conditions, producing paired measurements across all three strands.

### 5.1 RQ1 — Detection

Stress-detection performance is evaluated by F1-score, the primary metric throughout, given the class imbalance inherent in stress detection; accuracy and per-class confusion matrices are reported alongside for completeness. Success is defined at three levels. The floor is stress-detection F1 significantly above a majority-class baseline, the meaningful threshold for imbalanced binary classification. The substantive criterion is the transfer result: the classifier’s stress-detection F1 during the embodied interaction task relative to its calibration-context F1, which quantifies how much personalised detection survives the

move from controlled calibration to embodied MR — the question RQ1 actually asks. This calibration-to-interaction transfer gap is the same quantity whose stabilisation marks the exit of the Cycle 1 technical pilot (§4.5), evaluated here at full scale. For field context, calibration-context F1 is benchmarked against personalised physiological classification in comparable immersive work [12, 13], as a reference point for where the classifier sits rather than a pass-or-fail gate. Two established multimodal datasets serve as offline reference points, not pre-training sources: DEAP [35], with EEG and peripheral signals from 32 participants viewing 40 affective videos, and DREAMER [36], an EEG/ECG (electrocardiography) dataset from 23 participants viewing 18 film clips.

## 5.2 RQ2 — Response

Subjective experience is measured with three validated instruments administered post-session for each condition. The primary outcome is the Perceived Empathy of Technology Scale (PETS) [29], measuring perceived empathic responsiveness. Trust and acceptance are measured with the Almere model [39], validated for social agent technologies, and presence with the Igroup Presence Questionnaire (IPQ) [40]. Within-participant differences between conditions are tested with the Wilcoxon signed-rank test, appropriate for ordinal data at this sample size. A measurable difference would show that personalised detection translates, through adaptive response, into a distinguishable interaction; a null result would be informative in its own right, localising the limitation to the integration or to MR-specific factors — an outcome with direct precedent in Schmidmaier et al.’s finding of significant linguistic but no perceived-empathy effects [17].

## 5.3 RQ3 — Alignment

RQ3 analyses how the objective and subjective channels align across the same participants and interactions, treating alignment as a continuous empirical question rather than a binary verdict. Because the post-session instruments yield one value per condition, a continuous subjective signal is obtained by retrospective cued-recall: participants review a recording of their session and rate their experienced state along its timeline, a method that yields a continuous trajectory without interrupting the live interaction [26]. Four analyses, each addressing a distinct facet of alignment, relate this trajectory to the classifier’s stress output: Spearman correlation for the strength and direction of association; the intraclass correlation coefficient for absolute agreement, which covariation alone can mask; Cohen’s  $\kappa$  for categorical concordance once both channels are thresholded; and temporal cross-correlation for how closely the classifier tracks experienced state within a session. Convergence across these analyses is evidence that the system engages with the user’s experience rather than a population average [10]; divergence is a diagnostic finding locating where physiological inference and subjective experience part company, not a failure of the system. This treatment is warranted because the relationship between the channels is known to be variable: physiological and subjective measures of empathy are reported to correspond only erratically [11, 26], and Chang et al.’s Mixed Reality study found significant subjective differences between agent conditions alongside no significant physiological differences [19]. A known limitation is that controlled laboratory conditions may not fully represent natural MR deployment, which future field studies would need to address.

# 6 Project Management

**Scope and deliverables.** The project investigates whether personalised multimodal physiological classification can drive an empathic AI agent’s verbal responses in Mixed Reality, producing a measurable difference in interaction experience against a non-adaptive baseline. The deliverable is an operational prototype integrating physiological sensing (EEG, EDA, eye-gaze) with an adaptive LLM-based agent in an MR environment, with main-study results — classifier performance and subjective interaction measures — documented as a thesis. Commercial deployment and signal modalities beyond the EEG, EDA, and eye-gaze set are out of scope.

**Team and ownership.** The team divides work along the project’s three research strands, each with a designated owner reporting to the project lead. The lead holds overall design and integration. RQ1 (detection: physiological sensing and the personalised classifier), RQ2 (response: the adaptive empathic

agent), and RQ3 (alignment: convergence analysis of objective and subjective measures) each have a dedicated owner, with project coordination, scheduling, and risk tracking separately owned. The three research strands map directly onto the system’s detect–respond–align structure, so ownership follows the same spine as the methodology and evaluation.

**Timeline and milestones.** Table 3 summarises the work breakdown by phase across the 28-week two-semester window, and Figure 1 shows the corresponding timeline. The two validation phases are operationalised as the project’s design-based research cycles: the technical pilot (Cycle 1) refines the sensing pipeline and selects the classifier, and the process pilot (Cycle 2) refines the agent’s adaptive response before the main study. Phase-transition checkpoints occur at weeks 4, 12, 16, and 24, each requiring sign-off from the project lead before the next phase begins. The critical path runs design finalisation → ethics approval → technical pilot → main data collection, since delay in any of these blocks all subsequent work. Weekly team reviews run throughout, covering progress, risk status, and upcoming transitions.

Table 3: Work breakdown by phase.

| Phase                        | Key activities                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| Design and preparation       | Finalise experimental design; MR scenario design; recruitment materials                               |
| Ethics and build             | Ethics submission and review (parallel); build MR environment, agent pipeline, and sensing/classifier |
| Validation (DBR cycles)      | Cycle 1 technical pilot (sensing/classifier); Cycle 2 process pilot (interaction flow)                |
| Data collection and analysis | Main study across participants; classifier and subjective analysis                                    |
| Write-up                     | Thesis documentation                                                                                  |

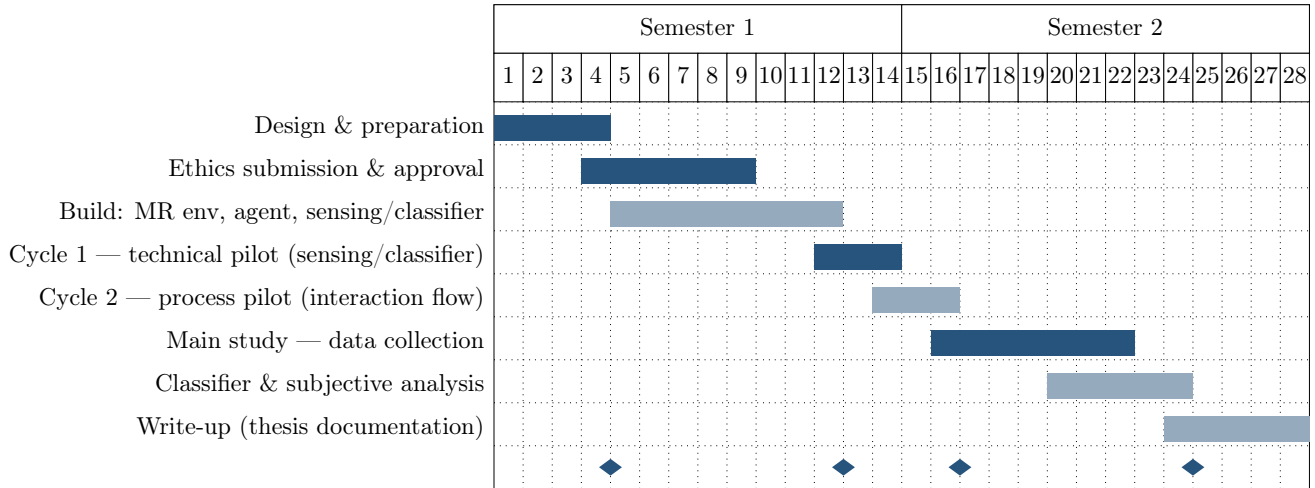


Figure 1: Project timeline across the 28-week, two-semester window. Darker bars mark the critical path (design finalisation → ethics approval → technical pilot → main data collection); diamonds mark lead sign-off gates at weeks 4, 12, 16, and 24. Ethics review runs in parallel with the build.

## Appendix: Generative AI Declaration

This work was primarily human-created. AI was used to make stylistic edits, such as changes to structure, wording, and clarity. AI was used to make content edits, such as changes to scope, information, and ideas. AI was prompted for its contributions, or AI assistance was enabled. AI-generated content was reviewed and approved. The following model(s) or application(s) were used: Claude Opus 4.6–4.8 (Anthropic) and Grammarly.

Generated via the AI Attribution Toolkit: AIA Ph SeCe Hin R — Claude Opus 4.6–4.8 v1.0

Detailed per-tool documentation follows in Tables 4 and 5 below.

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Table 4: Generative AI Tool Usage — Claude.

| Field                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. GenAI tool used                           | Claude (Anthropic), accessed via the Claude web interface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2. Purpose of GenAI tool for this assignment | <ul style="list-style-type: none"> <li>• Brainstorming research questions, methodology options, and section outlines</li> <li>• Drafting and refining content for the revised sections (methodology, evaluation, project management)</li> <li>• Rewriting, editing, and checking alignment against the brief across multiple paragraph-by-paragraph review passes</li> <li>• Citation reconciliation (resolving duplicate keys, verifying figures against source papers) and LaTeX formatting, including the <code>pgfgantt</code> timeline</li> </ul>                    |
| 3. Prompts or instructions given             | <p>Representative examples:</p> <ul style="list-style-type: none"> <li>• “Can you go through the draft and suggest any sections that require revision?”</li> <li>• “Can you help me tighten up this paragraph?”</li> <li>• “I don’t like saying ‘This matters because’ — the logic should explain what matters.”</li> </ul>                                                                                                                                                                                                                                               |
| 4. Amount of prompts or instructions given   | Approximately 100+ prompts across multiple sessions over several days.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5. Summary of GenAI output                   | Claude proposed structural decisions, refined draft sentences, organised the by-RQ methodology, generated LaTeX formatting including the <code>pgfgantt</code> timeline, and performed alignment audits identifying rubric gaps and inconsistencies between sections.                                                                                                                                                                                                                                                                                                     |
| 6. GenAI output verification                 | All content was verified against source materials by the team. Every cited paper had been read, and specific numeric claims (F1 and accuracy figures, sample sizes, effect sizes) were cross-checked against source PDFs. Several claims carried over from the A2 draft were found inaccurate on re-checking and were corrected or removed.                                                                                                                                                                                                                               |
| 7. GenAI output usage or modification        | The team reviewed every paragraph, rewriting for clarity and register, cutting jargon-dense phrasing, pushing back on suggestions that overclaimed, and restructuring sections flagged for logical inconsistencies across multiple review passes.                                                                                                                                                                                                                                                                                                                         |
| 8. Reflection on generated content and usage | <p><b>What worked well:</b> identifying logical inconsistencies between sections, catching brief-alignment gaps, verifying figures against sources, and tightening wordy phrasing.</p> <p><b>What did not work well:</b> Claude occasionally overclaimed about source papers, attaching citations to claims the cited work did not fully support, requiring the team to cross-check and correct.</p> <p><b>Value added:</b> the team directed, verified, and refined; Claude suggested and flagged, surfacing structural issues earlier than solo writing would have.</p> |

Table 5: Generative AI Tool Usage — Grammarly.

| Field                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. GenAI tool used                           | Grammarly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2. Purpose of GenAI tool for this assignment | Grammar correction and sentence-structure checking on paragraphs after initial drafting.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3. Prompts or instructions given             | No explicit prompts. Grammarly operates by analysing pasted or typed text and surfacing inline suggestions. Text was submitted by pasting draft paragraphs into the Grammarly editor.                                                                                                                                                                                                                                                                                                                |
| 4. Amount of prompts or instructions given   | Not applicable (non-prompt-based tool). Used across a small number of editing passes on selected paragraphs.                                                                                                                                                                                                                                                                                                                                                                                         |
| 5. Summary of GenAI output                   | Grammarly flagged potential grammatical errors, suggested wording alternatives, and highlighted sentences flagged as unclear or overly long.                                                                                                                                                                                                                                                                                                                                                         |
| 6. GenAI output verification                 | All suggestions were reviewed individually. Grammar corrections were accepted only where they were grammatically correct and preserved the intended meaning. Some suggestions (particularly structural rewrites) were rejected where they altered the meaning or register of the original sentence.                                                                                                                                                                                                  |
| 7. GenAI output usage or modification        | Accepted grammar and punctuation corrections were applied directly. Sentence-structure suggestions were used as prompts to reconsider phrasing, but the team typically wrote their own reformulations rather than accepting Grammarly’s suggested rewrites verbatim.                                                                                                                                                                                                                                 |
| 8. Reflection on generated content and usage | <p><b>What worked well:</b> Grammarly was useful as a secondary check for typos and minor grammatical issues easy to miss in self-editing.</p> <p><b>What did not work well:</b> Suggested rewrites sometimes reduced specificity or altered technical meaning, particularly in sentences containing domain-specific terminology.</p> <p><b>Value added:</b> Grammarly acted as a final pass for surface-level correctness, complementing the substantive drafting and editing done with Claude.</p> |